Hall Ticket Number:

MR-21

Code: B515CS3S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. V Semester Supplementary Examinations, June/July 2025

Software Engineering

(Common to CSE and IT)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	List the characteristics of Software contrasting the characteristics of hardware.	2	1	1	1,8
1. b)	Mention the Advantage and Disadvantages of Waterfall model.	2	2	1	1,8
1. c)	Define feasibility study. And list the types.	2	1	2	2
1. d)	Differentiate between Functional and Non-functional Requirements.	2	2	2	2
1. e)	Write the importance of generalization relationship.	2	1	3	2
1. f)	List the components of use case diagram.	2	2	3	1
1. g)	Differentiate verification and validation. Which type of testing address verification? Which type of testing address validation?	2	2	4	2
1. h)	What are the approaches of debugging?	2	1	4	1
1. i)	Define Quality. How can quality be quantified.	2	2	5	1
1. j)	Define the terms i) Measures ii) Metrics and iii) Indicators.	2	2	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain in detail about the Phases in Unified Process Model with neat diagram.	5	2	1	2,3
2. b)	Explain about the Umbrella Activities which support the software development process and discuss about their necessity in maintaining.	5	5	1	2,3
	OR				
2. c)	Assume that you are the technical manager of a software development organization. A Client approached you for a software solution the problems stated by the client have uncertainties which lead to loss if it not planned and solved which software development model you will suggest for this project –justify. Explain that model with its pros and cons and neat sketch	5	5	1	2,3
2. d)	Discuss Manager's myths about software development and their effect on the practitioner's performance as well as on the overall outcome.	5	2	1	2,3
3. a)	Describe Extreme Programming Values and Framework Activities in detail.	5	2	2	2,3
3. b)	Demonstrate on Requirement Validation Techniques.	5	3	2	2
	OR				
3. c)	Give an in-depth overview of the dynamic systems development method and feature-driven development method.	5	4	2	2,3
3. d)	Compare the scenarios that are interrelated with Agility and the Cost of Change.	5	4	2	2
4. a)	Generalize on the concept of User Interface design and list the characteristics of good user interface design.	5	6	3	3
4. b)	Discuss the basic design principles of class-based components.	5	2	3	2,3
	OR				
4. c)	Explain about software architecture design, with emphasize as fan in, fan-out, coupling, cohesion and factoring.	5	2	3	2,3

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4. d)	Describe the differences and interrelations between the UML sequence diagram and activity diagram. Use real-world examples to illustrate your answer.	5	4	3	3
5. a)	Apprise and analyze the purpose of system testing with a high level Explanation on all its types	5	4	4	4
5. b)	Write a brief on equivalence partitioning & boundary value analysis of black box testing	5	3	4	3
	OR				
5. c)	Analyze elaborately on unit testing. How do you develop test suites.	5	3	4	4
5. d)	Analyze about alpha and beta testing.	5	3	4	4
6. a)	Discuss in detail about the Attributes of Effective Software Metrics.	5	2	5	5
6. b)	Examine the activities associated with project process planning.	5	4	5	5,11
	OR				
6. c)	Discuss the importance of "Metrics for Source Code" in software quality.	5	2	5	3
6. d)	What are the different ways in which quality can be reviewed? Explain them.	5	2	5	5

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome